

Technical Dossier Package

Multi-Algorithm MAR Comparison — ASTM-WKXXXXXX-v1.0.0

Generated 2026-03-17

1. Submission Identity

Field	Value
Mode	Multi-algorithm comparison
Algorithms	ASD-iMAR, DLSC-iMAR, FS-iMAR, MBIR-iMAR, POCS-TV, Spectral MAR, iMAR
Standard reference	ASTM-WKXXXXXX-v1.0.0
TDP version	1.0.0
Generation date	2026-03-17
T1_AB anchor tier	PRESENT

2. Algorithm Comparison by Tier

2.1 Tier T1_AB — Large adult-body Co-Cr rod — mandatory 510(k) anchor tier

ellipse 85×60 mm | CoCr μ =4.20 cm[■] | blockage 41.5%

Algorithm	AUC_noMAR	AUC_MAR	Δ AUC	95% CI Δ AUC	Outcome
Spectral MAR	0.5375	0.5256	-0.0119	[-0.0363, +0.0112]	NON-INFERIOR
iMAR	0.5375	0.5231	-0.0144	[-0.0381, +0.0037]	NON-INFERIOR

Spectral MAR (NON-INFERIOR):

MAR implementation 'Spectral MAR' demonstrates NON-INFERIOR lesion detectability vs noMAR for T1_AB (Large adult-body Co-Cr rod – mandatory 510(k) anchor tier; 41% angular blockage, CoCr) with Δ AUC = -0.0119 (95% CI [-0.0363, +0.0112]; one-tailed Wilcoxon p = 0.0000).

iMAR (NON-INFERIOR):

MAR implementation 'iMAR' demonstrates NON-INFERIOR lesion detectability vs noMAR for T1_AB (Large adult-body Co-Cr rod – mandatory 510(k) anchor tier; 41% angular blockage, CoCr) with Δ AUC = -0.0144 (95% CI [-0.0381, +0.0037]; one-tailed Wilcoxon p = 0.0000).

2.2 Tier T2_SB — Small pediatric-body SS-316L rod — supplemental tier

ellipse 42×30 mm | SS316L μ =2.80 cm[■] | blockage 40.0%

Algorithm	AUC_noMAR	AUC_MAR	Δ AUC	95% CI Δ AUC	Outcome
ASD-iMAR	0.5700	0.5041	-0.0659	[-0.1137, -0.0359]	INDETERMINATE
DLSC-iMAR	0.5700	0.5156	-0.0544	[-0.1219, -0.0062]	INDETERMINATE
FS-iMAR	0.8019	0.5400	-0.2619	[-0.3331, -0.1925]	INDETERMINATE

MBIR-iMAR	0.5700	0.5144	-0.0556	[-0.0888, -0.0187]	INDETERMINATE
Spectral MAR	0.5700	0.6188	+0.0488	[+0.0037, +0.0981]	SUPERIOR
iMAR	0.8019	0.5394	-0.2625	[-0.3381, -0.1913]	INDETERMINATE

ASD-iMAR (INDETERMINATE):

MAR implementation 'ASD-iMAR' demonstrates INDETERMINATE lesion detectability vs noMAR for T2_SB (Small pediatric-body SS-316L rod – supplemental tier; 40% angular blockage, SS316L) with $\Delta AUC = -0.0659$ (95% CI [-0.1137, -0.0359]; one-tailed Wilcoxon $p = 0.0886$).

DLSC-iMAR (INDETERMINATE):

MAR implementation 'DLSC-iMAR' demonstrates INDETERMINATE lesion detectability vs noMAR for T2_SB (Small pediatric-body SS-316L rod – supplemental tier; 40% angular blockage, SS316L) with $\Delta AUC = -0.0544$ (95% CI [-0.1219, -0.0062]; one-tailed Wilcoxon $p = 0.5920$).

FS-iMAR (INDETERMINATE):

MAR implementation 'FS-iMAR' demonstrates INDETERMINATE lesion detectability vs noMAR for T2_SB (Small pediatric-body SS-316L rod – supplemental tier; 40% angular blockage, SS316L) with $\Delta AUC = -0.2619$ (95% CI [-0.3331, -0.1925]; one-tailed Wilcoxon $p = 0.0013$).

MBIR-iMAR (INDETERMINATE):

MAR implementation 'MBIR-iMAR' demonstrates INDETERMINATE lesion detectability vs noMAR for T2_SB (Small pediatric-body SS-316L rod – supplemental tier; 40% angular blockage, SS316L) with $\Delta AUC = -0.0556$ (95% CI [-0.0888, -0.0187]; one-tailed Wilcoxon $p = 0.0000$).

Spectral MAR (SUPERIOR):

MAR implementation 'Spectral MAR' demonstrates SUPERIOR lesion detectability vs noMAR for T2_SB (Small pediatric-body SS-316L rod – supplemental tier; 40% angular blockage, SS316L) with $\Delta AUC = +0.0488$ (95% CI [+0.0037, +0.0981]; one-tailed Wilcoxon $p = 0.0000$).

iMAR (INDETERMINATE):

MAR implementation 'iMAR' demonstrates INDETERMINATE lesion detectability vs noMAR for T2_SB (Small pediatric-body SS-316L rod – supplemental tier; 40% angular blockage, SS316L) with $\Delta AUC = -0.2625$ (95% CI [-0.3381, -0.1913]; one-tailed Wilcoxon $p = 0.0001$).

2.3 Tier T3_HEAD — Circular head phantom Ti-6Al-4V rod — supplemental tier

circle $r=100$ mm | Ti6Al4V $\mu=1.50$ cm⁻¹ | blockage 16.1%

Algorithm	AUC_noMAR	AUC_MAR	ΔAUC	95% CI ΔAUC	Outcome
FS-iMAR	0.5331	0.5525	+0.0194	[-0.0238, +0.0581]	NON-INFERIOR
POCS-TV	0.5331	0.5138	-0.0194	[-0.0550, +0.0062]	INDETERMINATE
Spectral MAR	0.5331	0.5050	-0.0281	[-0.0644, -0.0025]	INDETERMINATE
iMAR	0.5331	0.5262	-0.0069	[-0.0475, +0.0275]	NON-INFERIOR

FS-iMAR (NON-INFERIOR):

MAR implementation 'FS-iMAR' demonstrates NON-INFERIOR lesion detectability vs noMAR for T3_HEAD (Circular head phantom Ti-6Al-4V rod – supplemental tier; 16% angular blockage, Ti6Al4V) with $\Delta AUC = +0.0194$ (95% CI [-0.0238, +0.0581]; one-tailed Wilcoxon $p = 0.0014$).

POCS-TV (INDETERMINATE):

MAR implementation 'POCS-TV' demonstrates INDETERMINATE lesion detectability vs noMAR for T3_HEAD (Circular head phantom Ti-6Al-4V rod – supplemental tier; 16% angular blockage,

Ti6Al4V) with $\Delta\text{AUC} = -0.0194$ (95% CI $[-0.0550, +0.0062]$; one-tailed Wilcoxon $p = 0.0155$).

Spectral MAR (INDETERMINATE):

MAR implementation 'Spectral MAR' demonstrates INDETERMINATE lesion detectability vs noMAR for T3_HEAD (Circular head phantom Ti-6Al-4V rod – supplemental tier; 16% angular blockage, Ti6Al4V) with $\Delta\text{AUC} = -0.0281$ (95% CI $[-0.0644, -0.0025]$; one-tailed Wilcoxon $p = 0.0010$).

iMAR (NON-INFERIOR):

MAR implementation 'iMAR' demonstrates NON-INFERIOR lesion detectability vs noMAR for T3_HEAD (Circular head phantom Ti-6Al-4V rod – supplemental tier; 16% angular blockage, Ti6Al4V) with $\Delta\text{AUC} = -0.0069$ (95% CI $[-0.0475, +0.0275]$; one-tailed Wilcoxon $p = 0.0063$).

3. Methodology Traceability

Parameter	Value
Observer	2D Channelized Hotelling Observer (CHO)
Channels	10 Laguerre-Gauss (tier-specific a)
Covariance	LA condition only (§A1.5.2(a))
Regularization	Tikhonov $\lambda = 0.01 \times \text{trace}(K) / 10$
Hold Out	Leave-One-Out (LOO) with observer re-fit per fold
Bootstrap	1000 resamples, observation-level, on LOO scores
Wilcoxon	One-tailed paired Wilcoxon on $d[i] = s_LP[i] - s_LA[i]$
Estimation Bias	50/50 2-fold CV
Metric	Mann-Whitney AUC (mid-rank tie correction)
Note 3D	3D integration across z PROHIBITED (§A1.5.3)